Notes: Malmendier and Nagel (QJE, 2011)

Depression Babies: Do Macroeconomic Experiences Affect Risk Taking?

Contents

1 Big picture

- **Question.** Do people who live through different macroeconomic histories end up taking different levels of financial risk later in life?
- **Motivating idea.** The paper starts from the popular intuition of "Depression babies": people who experienced the Great Depression are often thought to remain more cautious investors for decades.
- **Core challenge.** Age, calendar time, and cohort are tightly linked, so it is hard to tell whether low stock market participation among some generations comes from age, from the current aggregate environment, or from the specific return history they personally experienced.
- **Main methodological idea.** The paper does *not* estimate unrestricted cohort effects. Instead, it constructs for each household an *experienced return*: a weighted average of the stock or bond returns realized over that household head's lifetime. The data then reveal both:
  - how much experienced returns matter for current risk taking, and
  - how heavily recent versus distant past returns are weighted.
- **Main empirical setting.** The core evidence comes from repeated cross-sections of U.S. households in the Survey of Consumer Finances (SCF), 1960–2007, merged with long-run real stock and bond return series going back to the nineteenth century.
- **Main findings.**
  - Higher experienced stock returns make households more willing to take financial risk, more likely to own stocks, and more willing to allocate a larger share of liquid assets to stocks.
  - Higher experienced bond returns make households more likely to participate in the bond market.
  - Recent experiences receive more weight than distant ones, but experiences from many years ago still matter.
  - Higher experienced stock returns also predict more optimistic expectations about future stock returns.
- **Economic magnitudes.**
  - Moving from the 10th to the 90th percentile of experienced stock returns raises stock market participation by about **10.2 percentage points**.
  - The same change lowers the probability of being in the lowest self-reported financial-risk-tolerance category by about **10.3 percentage points**.
  - Among stockholders, it raises the fraction of liquid assets invested in stocks by about **7.9 percentage points**.
- **Takeaway.** Financial risk taking is not pinned down only by stable preferences and publicly available historical data. Personal macroeconomic experience shapes beliefs and behavior in a persistent, but recency-weighted, way.

2 Data and measurement (key objects)

2.1 Household microdata: Survey of Consumer Finances

- The core household data come from the **Survey of Consumer Finances (SCF)**.
- The sample has two pieces:
  - the modern SCF from **1983–2007**, available every three years;
  - the precursor SCF surveys from **1960, 1962, 1963, 1964, 1967, 1968, 1969, 1970, 1971, and 1977**.
- The authors start in 1960 because earlier surveys do not measure age precisely enough to construct lifetime return histories cleanly.
- Sample restrictions:
  - household head age between **25 and 74**;
  - variables weighted with SCF sample weights so the estimates represent the U.S. population.
- The SCF also oversamples rich households in later years, but the use of survey weights corrects this.

2.2 Return data and experienced returns

- Real stock returns come from the **S&P index** series, going back to **1871**.
- Real bond returns come from a total return index of **10-year U.S. Treasury bonds**, combined with CPI inflation to convert to real returns.
- For every household in every survey year, the authors reconstruct the annual sequence of stock and bond returns observed since the birth year of the household head.
- These lifetime histories are then compressed into a single summary statistic—the **experienced return**—using an estimated weighting function.

2.3 Risk-taking measures

- **Elicited risk tolerance** (available in 1983 and 1989–2007): respondents choose among four categories ranging from no financial risk to substantial financial risk.
- **Stock market participation**: indicator for whether the household owns any stocks, including stock mutual funds and, in the baseline, stocks in retirement accounts.
- **Bond market participation**: indicator for whether the household owns long-term bonds.
- **Fraction of liquid assets invested in stocks**: among stockholders, the share of liquid assets allocated to stocks.
- Liquid assets are defined as stocks, bonds, cash, and short-term instruments.

2.4 Key descriptive facts

- In the full sample of **51,204** household observations:
  - mean stock market participation is **34.2%**;
  - mean bond market participation is **37.6%**;
  - mean elicited risk tolerance is **1.826** on the 1–4 scale.
- Median liquid assets are only **$6,382** in the full sample, but much higher among participants:
  - **$51,567** for stock market participants;
  - **$31,842** for bond market participants.
- Mean experienced real returns (using an illustrative weighting parameter $\lambda = 1.50$) are:
  - stock: **9.1%**;
  - bond: **1.8%**.
- Their cross-sectional variation is economically meaningful:
  - experienced stock returns range from **6.2%** at the 10th percentile to **11.9%** at the 90th percentile;
  - experienced bond returns range from about **-0.2%** to **5.1%**.

2.5 Controls and unobservables

- All main specifications include:
  - **age dummies**,
  - **year dummies**,
  - income controls,
  - liquid-asset controls,
  - household characteristics such as marital status, retirement, race, education, children, and pension-plan indicators.
- This is crucial because the paper wants to isolate how *different lifetime macro histories* matter, not merely age or aggregate stock market conditions.

3 Models and empirical specifications (all equations + interpretation)

3.1 The core experience measure

For household i in survey year t, the experienced return is

$$A_{it}(\lambda) = \sum_{k=1}^{age_{it}-1} w_{it}(k, \lambda) R_{t-k},$$

with weights

$$w_{it}(k,\lambda) = \frac{(age_{it}-k)^{\lambda}}{\sum_{j=1}^{age_{it}-1}(age_{it}-j)^{\lambda}}.$$

- R_{t-k} is the realized return k years ago.
- age_{it} is the household head's age.
- λ determines how strongly recent returns are emphasized.

Interpretation.

- If $\lambda = 0$, all past returns since birth receive equal weight.
- If $\lambda > 0$, recent returns get more weight than distant returns.
- If $\lambda < 0$, early-life returns get more weight than recent returns.

The key empirical question is whether the data choose $\lambda > 0$ and whether $A_{it}(\lambda)$ significantly predicts current risk taking.

3.2 Generic specification

The paper's reduced-form backbone is

$$y_{it} = \alpha + \beta A_{it}(\lambda) + \gamma' x_{it} + \varepsilon_{it},$$

where:

- y_{it} is a measure of current risk taking,
- $A_{it}(\lambda)$ is experienced return,
- x_{it} contains age, year, wealth, income, and household controls.

Interpretation.

- β tells us whether better lifetime return experiences increase risk taking.
- λ tells us how fast the influence of past experiences decays.

3.3 Ordered probit for elicited risk tolerance

For self-reported risk tolerance, the paper estimates

$$\Pr(y_{it} \le j \mid x_{it}, A_{it}(\lambda)) = \Phi(\alpha_j - \beta A_{it}(\lambda) - \gamma' x_{it}), \qquad j \in \{1,2,3,4\}.$$

- The dependent variable is the 1–4 risk-tolerance category.
- The model is ordered probit because the categories are ordinal.

Main result from Table II.

- With full controls, the experienced stock return coefficient is $\beta = 6.692$ (s.e. **1.180**).

- The weighting parameter is $\lambda = 1.433$ (s.e. **0.280**).
- Moving from the 10th to the 90th percentile of experienced stock returns lowers the probability of being in the lowest risk-tolerance category by **10.3 pp**.

Interpretation.

- The positive effect of experienced stock returns is strong and economically meaningful.
- The estimate $\lambda \approx 1.4$ means recent returns matter more, but even returns from decades ago still affect today's reported willingness to take risk.

3.4 Probit for stock and bond market participation

For participation, the paper estimates

$$\Pr(y_{it} = 1 \mid x_{it}, A_{it}(\lambda)) = \Phi(\alpha + \beta A_{it}(\lambda) + \gamma' x_{it}).$$

- In the stock regression, $A_{it}(\lambda)$ is experienced **stock** return.
- In the bond regression, $A_{it}(\lambda)$ is experienced **bond** return.

Main stock-participation result (Table III, column ii).

- $\beta =$ **10.627** (s.e. **1.403**),
- $\lambda =$ **1.325** (s.e. **0.209**),
- 10th-to-90th percentile change in experienced stock returns raises stock participation by **10.2 pp**.

Main bond-participation result (Table III, column iv).

- $\beta =$ **8.433** (s.e. **1.680**),
- $\lambda =$ **1.282** (s.e. **0.367**),
- 10th-to-90th percentile change in experienced bond returns raises bond participation by **11.4 pp**.

Interpretation.

- Experience effects appear in both stock and bond participation.
- The estimated weighting patterns are very similar across asset classes.
- The stock result is especially robust; the bond result is somewhat less stable in subsamples because experienced bond returns vary less flexibly across cohorts and time.

3.5 Nonlinear least squares for the stock share

For the fraction of liquid assets invested in stocks, among stockholders, the paper estimates

$$y_{it} = \alpha + \beta A_{it}(\lambda) + \gamma' x_{it} + \varepsilon_{it},$$

with nonlinear least squares because $A_{it}(\lambda)$ is nonlinear in λ.

Main result (Table IV, column ii).

- $\beta =$ **1.668** (s.e. **0.395**),
- $\lambda =$ **1.166** (s.e. **0.299**),
- a 10th-to-90th percentile increase in experienced stock returns raises the stock share by about **7.9 pp**.

The paper also estimates the same equation using **experienced excess stock returns** (stock minus bond returns).

Excess-return result (Table IV, column iv).

- $\beta =$ **1.734** (s.e. **0.423**),
- $\lambda =$ **1.831** (s.e. **0.428**),
- the 10th-to-90th percentile shift raises the stock share by about **13.4 pp**.

Interpretation.

- This is especially striking because the literature often finds that, conditional on participation, few household characteristics strongly explain the risky share.
- Here, lifetime return experience explains a large part of the heterogeneity in how aggressively participants invest in stocks.

3.6 Using stock and bond returns jointly

To distinguish belief updating about particular asset classes from more general shifts in risk aversion, the paper includes experienced stock and bond returns *together* in the same specification.

Main results (Table V).

- **Stock participation**: experienced stock returns remain strongly positive (10.564), while experienced bond returns are close to zero (0.944).
- **Bond participation**: experienced bond returns remain strongly positive (8.562), while experienced stock returns are small and negative (−1.149).
- **Stock share**: experienced stock returns are positive (1.765), while experienced bond returns are negative (−0.748).
- **Bond share**: experienced bond returns are positive (0.970), while experienced stock returns are negative (−1.381).

Interpretation.

- The results line up more naturally with an **asset-specific beliefs** channel than with a simple one-dimensional risk-aversion story.
- If people become optimistic specifically about the asset class in which they had good experiences, stock experiences should predict stock demand more than bond demand, and vice versa. That

is exactly what the paper finds.

3.7 Expectations regressions

To study the beliefs channel directly, the paper turns to the **UBS/Gallup survey** on stock return expectations from 1998–2007.

It estimates regressions of the form

$$\text{Expected return}_{it} = \alpha + \beta A_{it}(\lambda) + \gamma' x_{it} + \varepsilon_{it},$$

where λ is fixed at the values estimated in the baseline SCF specifications.

Main results (Table VI).

- For expected stock market return over the next 12 months, the experienced stock return coefficient is about **0.53–0.62**.
- For the expected return on the respondent's own portfolio, it is about **0.58–0.65**.

Interpretation.

- A **1 percentage point** increase in experienced stock returns predicts roughly a **0.5–0.6 percentage point** increase in expected future stock returns.
- This is hard to reconcile with a pure stable-preferences story and strongly supports a **beliefs channel**. It does not rule out preference changes, but it shows beliefs do a substantial part of the work.

3.8 Aggregate perspective

The micro estimates imply aggregate consequences. The paper aggregates experienced stock returns across age groups using age-group liquid-asset weights and compares the resulting series to the stock market **P/E ratio**.

Interpretation.

- Periods with high average experienced stock returns coincide with periods of high market valuation, such as the **1960s** and **1990s**.
- Periods with low average experienced returns coincide with low valuation periods, such as the **1940s** and early **1980s**.
- This is not mechanical: the weighting parameter λ was estimated from *micro* cross-sectional behavior, not chosen to fit aggregate asset prices.

4 Identification and empirical credibility (what makes the estimates convincing)

4.1 How the paper handles the age–period–cohort problem

- Standard cohort-dummy regressions run into the age–time–cohort collinearity problem.

- This paper avoids that trap by not asking whether an arbitrary cohort fixed effect matters. Instead, it tests a *specific theory*: individuals with better lifetime return experiences should take more risk.
- Since experienced returns vary not only across cohorts but also *within* cohorts over time as new returns are realized, the paper can identify the effect even with age and year dummies included.

4.2 Why reverse causality is less of a problem than it first appears

- One concern is that current aggregate risk tolerance may move stock prices, making past returns and current risk taking mechanically correlated.
- The paper's solution is to include **year dummies**. That absorbs all aggregate time effects, including market-wide shifts in risk aversion, market clearing, and aggregate wealth fluctuations.
- Identification then comes from **cross-sectional differences** in risk taking associated with **cross-sectional differences** in lifetime return histories.

4.3 Why simple wealth effects are unlikely to explain the main results

- Experienced returns are naturally correlated with current wealth, so one may worry that the paper is just picking up wealth-dependent risk aversion.
- The specifications control flexibly for income and liquid assets.
- More importantly, the stock-versus-bond results are too asset-specific for a simple wealth story:
  - experienced stock returns matter mainly for stock risk taking,
  - experienced bond returns matter mainly for bond risk taking.

4.4 Beliefs versus preferences

- The paper cannot fully separate beliefs from preferences using the SCF alone.
- But the expectations evidence from UBS/Gallup strongly suggests that past returns affect **beliefs about future returns**.
- The joint stock-and-bond regressions also point in this direction because they imply **asset-specific optimism or pessimism**, not just a scalar shift in general risk tolerance.

4.5 Robustness checks

- The paper checks more flexible weighting functions and finds little evidence that nonmonotonic weighting is important.
- Excluding retirement assets leaves the core results largely unchanged.
- Shifting the starting point of lifetime experience to 10 years before or after birth also changes little.

- Adding cohort dummies leaves most results similar, though the bond-participation results become much less precise.
- Experienced volatility does not have a consistent significant effect once experienced returns are included.

4.6 Main limitation

- The paper measures experience using *public market returns*, not the actual personal returns or information sets of households.
- This is sensible empirically, but it means the estimated experience variable is a proxy for the true experiences and narratives households internalized.

5 Tables and figures

5.1 Figure I and Table I: raw pattern and descriptive facts

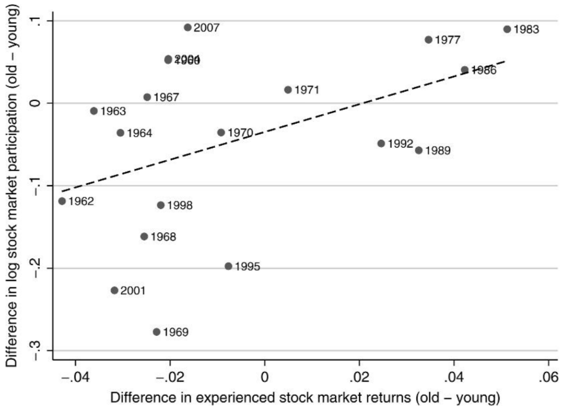

FIGURE I

Differences in Log Stock Market Participation Rates of Old and Young Individuals Plotted against Differences in Experienced Stock Market Returns

Stock market participation rates are the fraction of households who invest in stocks, including stock mutual funds and stocks held in retirement accounts. The vertical axis shows the difference in the log participation rates of old (household head age > 60 years) minus young (household head age ≤ 40 years) households. The horizontal axis shows the average real stock market return (S&P 500 index) over the prior 50 years (as proxy for the return experienced by old households) minus the return over the prior 20 years (as proxy for the return experienced by young households). The years refer to the respective SCF survey waves. Observations are weighted with SCF sample weights. The 1960 and 2004 observations overlap in the plot.

TABLE I

SUMMARY STATISTICS

	10th pct	Median	90th pct	Mean	Std. dev.	#Obs.
Panel A: All households						
Liquid assets	0	6,382	144,639	80,248	652,585	51,204
Income	11,883	43,419	105,339	60,511	178,088	51,204
Experienced real stock return (λ = 1.50)	0.062	0.092	0.119	0.091	0.022	51,204
Experienced real bond return (λ = 1.50)	−0.002	0.006	0.051	0.018	0.022	51,204
Stock market participation	0	0	1	0.342	0.474	51,204
Bond market participation	0	0	1	0.376	0.484	49,866
Elicited risk tolerance (1983–2007)	1	2	3	1.826	0.841	28,732
Panel B: Stock market participants						
Liquid assets	4,974	51,567	409,723	210,070	1,108,027	22,541
Income	28,339	67,799	164,596	99,759	297,869	22,541
Bond market participation	0	1	1	0.681	0.466	22,293
Fraction of liquid assets in stocks	0.071	0.451	0.900	0.466	0.295	21,669
Elicited risk tolerance (1983–2007)	1	2	3	2.103	0.801	17,049
Panel C: Bond market participants						
Liquid assets	2,332	31,842	322,119	169,981	1,027,384	22,553
Income	26,113	61,926	143,552	89,119	275,068	22,553
Stock market participation	0	1	1	0.628	0.483	22,553
Fraction of liquid assets in stocks	0	0.157	0.740	0.266	0.294	21,556
Elicited risk tolerance (1983–2007)	1	2	3	2.010	0.804	15,954

Notes. The sample period is 1960–2007. Stock returns and bond returns are real returns, deflated with Consumer Price Index (CPI) inflation rates. Wealth and income variables are deflated by the CPI into September 2007 dollars. Observations are weighted by SCF sample weights. The bond market participant sample in Panel C excludes the 1964 survey in which bond market participation information is not available. Elicited risk tolerance is not available in the 1986 survey pct, percentile.

Read:

- Figure I is the raw motivating figure. It plots the difference in log stock market participation between older and younger households against the difference in their experienced stock returns.
- The raw pattern already matches the paper's hypothesis: when the young have worse recent return experience than the old, they participate less in the stock market relative to the old.

- Table I shows the baseline descriptive facts:
  - stock participation is **34.2%**;
  - bond participation is **37.6%**;
  - stockholders are much wealthier than the average household;
  - experienced stock returns vary by about **5.7 percentage points** between the 10th and 90th percentiles.

5.2 Figure II and Table II: weighting function and elicited risk tolerance

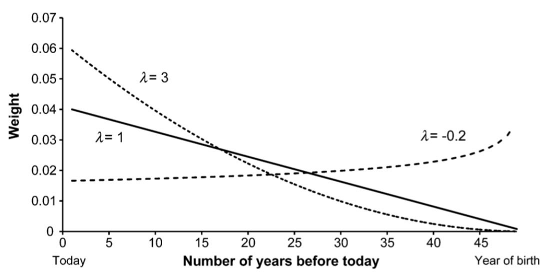

FIGURE II
Weights on Experienced Returns for Different Values of λ for a 50-year-old Household Head

TABLE II
ELICITED RISK TOLERANCE

	(i)	(ii)
Experienced stock return coefficient β	4.533	6.692
	(1.106)	(1.180)
Weighting parameter λ	1.834	1.433
	(0.452)	(0.280)
Income controls	Yes	Yes
Liquid assets controls	–	Yes
Household characteristics	Yes	Yes
Age dummies	Yes	Yes
Year dummies	Yes	Yes
Average of fitted prob. at 90th pct. minus fitted prob. at 10th pct. of experienced stock return		
Risk tolerance = 1 (low)	−0.088	−0.103
[unconditional freq. = 0.413]	(0.020)	(0.016)
Risk tolerance = 2	0.025	0.029
[unconditional freq. = 0.391]	(0.001)	(0.003)
Risk tolerance = 3	0.040	0.047
[unconditional freq. = 0.153]	(0.011)	(0.010)
Risk tolerance = 4 (high)	0.023	0.027
[unconditional freq. = 0.043]	(0.009)	(0.009)
#Obs.	28,571	28,571
Pseudo R^2	0.09	0.10

Notes. Ordered probit model estimated with maximum likelihood. The sample period runs from 1983 to 2007 and excludes the 1986 survey (elicited risk tolerance not available). The experienced stock return is calculated from the real return on the S&P 500 index. Liquid assets controls are log liquid assets and log liquid assets squared, both interacted with year dummies to allow for year-specific slopes. Household characteristics include the number of children and number of children squared, the percentage of liquid assets invested in defined contribution pension plans and IRAs, as well as dummies for marital status, retirement, race, education, for having a defined benefit pension plan, and for having a defined contribution pension plan or IRA. Observations are weighted with SCF sample weights. Standard errors, shown in parentheses, are robust to misspecification of the likelihood function and adjusted for multiple imputation pct, percentile.

Read:

- Figure II explains the entire empirical design. It shows how different values of λ imply very different lifetime weighting schemes for a 50-year-old.
- Table II shows the main risk-tolerance result: higher experienced stock returns make households less likely to report the lowest risk tolerance and more likely to report higher categories.
- The estimate $\lambda = 1.433$ implies a clear recency bias but not a short memory; decades-old experiences still matter.

5.3 Table III and Figure III: participation decisions and counterfactuals

Read:

- Table III shows that experienced stock returns predict stock participation, and experienced bond returns predict bond participation, both with large magnitudes.

TABLE III

STOCK AND BOND MARKET PARTICIPATION

	Experienced stock returns and stock market participation		Experienced bond returns and bond market participation	
	(i)	(ii)	(iii)	(iv)
Experienced return coefficient β	6.963	10.627	10.809	8.433
	(1.149)	(1.403)	(1.446)	(1.680)
Weighting parameter λ	1.924	1.325	1.176	1.282
	(0.267)	(0.209)	(0.218)	(0.367)
Income controls	Yes	Yes	Yes	Yes
Liquid assets controls	–	Yes	–	Yes
Household characteristics	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes
Average of fitted prob. at 90th pctile. minus	0.092	0.102	0.161	0.114
fitted prob. at 10th pct. of experienced return	(0.019)	(0.017)	(0.022)	(0.023)
#Obs.	51,013	51,013	49,678	49,678
Pseudo R^2	0.40	0.49	0.27	0.33

Notes. Probit model estimated with maximum likelihood. The sample period runs from 1960 to 2007 (excluding 1964 in the case of bond market participation). The experienced stock return in columns (i) and (ii) is calculated from the real return on the S&P 500 index. The experienced bond return in columns (iii) and (iv) is calculated from the real return on long-term U.S. Treasury bonds. Liquid assets controls are log liquid assets and log liquid assets squared, both interacted with year dummies to allow for year-specific slopes. Household characteristics include the number of children and number of children squared, the percentage of liquid assets invested in defined contribution pension plans and IRAs, as well as dummies for marital status, retirement, race, education, having a defined benefit pension plan, and having a defined contribution pension plan or IRA. Observations are weighted with SCF sample weights. Standard errors, shown in parentheses, are robust to misspecification of the likelihood function and adjusted for multiple imputation.

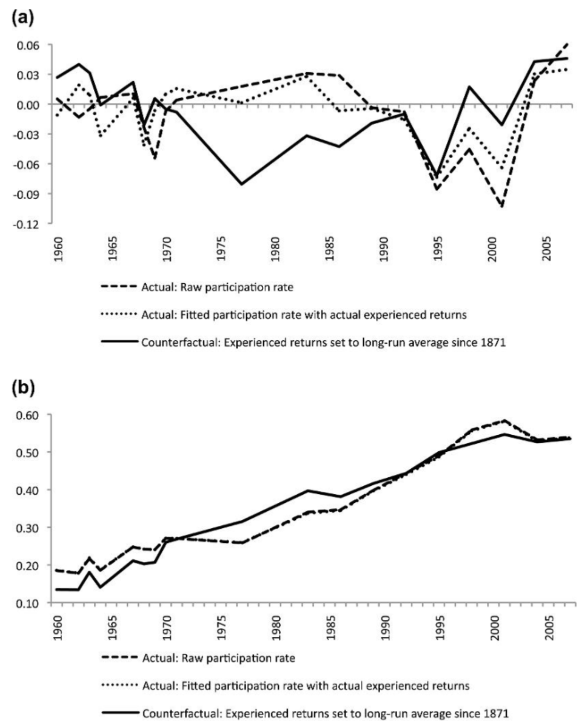

FIGURE III

Counterfactual Stock Market Participation Rates

Panel A shows the difference in average fitted stock market participation probabilities between young (age ≤ 40) and old (age > 60) (old minus young) from the probit model of Table III, column (ii), (dashed line), compared with the counterfactual difference in average fitted probabilities when the experienced stock market return is set to the average annual return since 1871 until the year prior to the survey year (solid line), and the actual participation rate in the raw data (dotted line). Panel B plots the average fitted and actual participation probabilities in the whole sample. Observations are weighted with SCF sample weights.

- The participation effects are around **10–11 percentage points** when moving from the 10th to the 90th percentile of experience.
- Figure III translates the regression into economically intuitive counterfactuals. When the paper replaces lifetime experiences with the long-run average return since 1871, it can almost reverse the young-versus-old participation gap in the early 1980s and eliminate much of the gap in the late 1990s.
- The same figure also shows that aggregate stock participation in the whole sample can differ from this counterfactual by up to about **7 percentage points**.

5.4 Table IV: portfolio allocation conditional on participation

Read:

- Table IV is important because it shows the effect is not only about crossing the fixed-cost margin of entering the stock market.
- Even among participants, better stock return experiences lead households to hold much more of their liquid wealth in stocks.
- The baseline effect is about **7.9 percentage points**; with excess stock-over-bond returns it is even larger, about **13.4 percentage points**.
- This is unusually strong relative to the standard household-portfolio literature, where the risky share is often hard to explain with demographics alone.

TABLE IV
FRACTION OF LIQUID ASSETS INVESTED IN STOCKS

	Experienced stock returns		Experienced excess returns of stocks over bonds	
	(i)	(ii)	(iii)	(iv)
Experienced return coefficient β	0.972	1.668	1.076	1.734
	(0.362)	(0.395)	(0.391)	(0.423)
Weighting parameter λ	1.299	1.166	1.736	1.831
	(0.520)	(0.299)	(0.622)	(0.428)
Income controls	Yes	Yes	Yes	Yes
Liquid assets controls	–	Yes	–	Yes
Household characteristics	Yes	Yes	Yes	Yes
Age dummies	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes
Average of fitted allocation stocks at 90th pctile. minus fitted allocation at 10th pctile. of experienced return	0.048	0.079	0.083	0.134
	(0.018)	(0.019)	(0.030)	(0.033)
#Obs.	21,611	21,611	21,611	21,611
R^2	0.12	0.16	0.12	0.16

Notes. Model estimated with nonlinear least squares on the sample of stock market participants. The sample period runs from 1960 to 2007, excluding the 1971 survey (allocation to stocks not available). Experienced stock returns are calculated from the real return on the S&P 500 index, experienced excess return from the return on the S&P 500 index minus the return on long-term U.S. Treasury bonds. Liquid assets controls are log liquid assets and log liquid assets squared, both interacted with year dummies to allow for year-specific slopes. Household characteristics include the number of children and number of children squared, the percentage of liquid assets invested in defined contribution pension plans and IRAs, as well as dummies for marital status, retirement, race, education, having a defined benefit pension plan, and having a defined contribution pension plan or IRA. Observations are weighted with SCF sample weights. Standard errors, shown in parentheses, are robust to heteroskedasticity and adjusted for multiple imputation.

TABLE V
USING STOCK AND BOND RETURNS JOINTLY

Dependent variable	Elicited risk tolerance	Stock market participation	Bond market participation	Fraction of liquid assets in stocks	Fraction of liquid assets in bonds
Sample	Full	Full	Full	Stock market participants	Bond market participants
	(i)	(ii)	(iii)	(iv)	(v)
Experienced stock return coeff. β_{stock}	4.670	10.564	–1.149	1.765	–1.381
	(2.042)	(1.405)	(1.244)	(0.410)	(0.361)
Weighting parameter for stocks λ_{stock}	1.433	1.325	1.325	1.166	1.166
	[fixed]	[fixed]	[fixed]	[fixed]	[fixed]
Average of fitted values at 90th pct. minus fitted values at 10th pct. of experienced stock return…	0.072	0.101	–0.015	0.103	–0.085
	(0.029)	(0.016)	(0.016)	(0.024)	(0.022)
…where fitted value refers to	probability of not being in lowest risk tolerance category	probability of stock market participation	probability of bond market participation	fraction of liquid assets in stocks	fraction of liquid assets in bonds

TABLE V
(CONTINUED)

Dependent variable	Elicited risk tolerance	Stock market participation	Bond market participation	Fraction of liquid assets in stocks	Fraction of liquid assets in bonds
Sample	Full	Full	Full	Stock market participants	Bond market participants
	(i)	(ii)	(iii)	(iv)	(v)
Experienced bond return coeff. β_{bond}	5.413	0.944	8.562	–0.748	0.970
	(4.420)	(1.904)	(1.579)	(0.572)	(0.487)
Weighting parameter for bonds λ_{bond}	1.282	1.282	1.282	1.282	1.282
	[fixed]	[fixed]	[fixed]	[fixed]	[fixed]
Average of fitted values at 90th pctile. minus fitted values at 10th pctile. of experienced bond return…	0.101	0.009	0.115	–0.036	0.046
	(0.079)	(0.018)	(0.021)	(0.028)	(0.023)
…where fitted value refers to	probability of not being in lowest risk tolerance category	probability of stock market participation	probability of bond market participation	fraction of liquid assets in stocks	fraction of liquid assets in bonds

Notes. Models and controls as in Table II, column (ii), Table III, columns (ii) and (iv), and Table IV, column (ii), but with experienced real stock and bond returns jointly included as explanatory variables and with the λ parameters fixed at the point estimates obtained in those earlier regressions. The experienced stock return is calculated from the real return on the S&P 500 index. The experienced bond return is calculated from the real return on long-term U.S. Treasury bonds. Observations are weighted with SCF sample weights. Standard errors, shown in parentheses, are robust to misspecification of the likelihood function/heteroskedasticity and adjusted for multiple imputation.

5.5 Table V: stock and bond returns jointly

Read:

- This table is the cleanest evidence against a generic wealth-effect explanation.
- Stock experiences matter mainly for stock decisions, and bond experiences matter mainly for bond decisions.
- That pattern is much more consistent with households learning or extrapolating about *particular asset classes* than with a single common shift in background risk tolerance.

5.6 Table VI and Figure IV: beliefs and aggregate implications

Read:

TABLE VI

EXPLAINING STOCK MARKET RETURN EXPECTATIONS WITH EXPERIENCED RETURNS

	Expected stock market return over the next 12 months (1998–2003)		Expected return on own portfolio over the next 12 months (1998–2007)	
	(i)	(ii)	(iii)	(iv)
Experienced stock return coefficient β	0.616	0.528	0.654	0.584
	(0.311)	(0.275)	(0.189)	(0.164)
Weighting parameter λ	1.166	1.325	1.166	1.325
	[fixed]	[fixed]	[fixed]	[fixed]
Age dummies	Yes	Yes	Yes	Yes
Year-month dummies	Yes	Yes	Yes	Yes
#Obs.	40,145	40,145	72,631	72,631
R^2	0.08	0.08	0.06	0.06

Notes. Dependent variable is the expected return over the next 12 months reported by individual respondents in the UBS/Gallup survey. Estimation is done with least squares, weighted with sample weights. The λ parameters are fixed at the point estimates obtained in column (ii) of Tables III and IV. The experienced stock return is calculated from the real return on the S&P 500 index. Standard errors, shown in parentheses, are robust to heteroskedasticity.

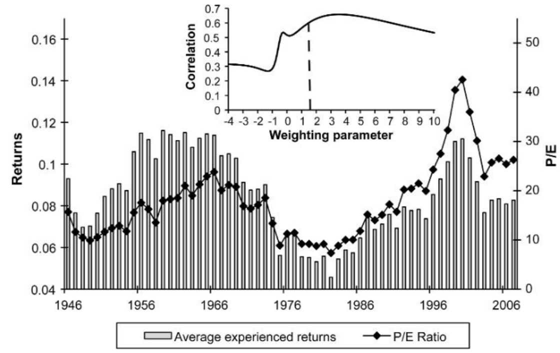

FIGURE IV

Average Experienced Real Stock Returns (λ = 1.50) and P/E Ratio of the S&P 500 Index 1946–2007

The P/E ratio is an updated series from Shiller (2005), which is calculated with a 10-year moving average of trailing earnings of firms in the S&P 500 index in the denominator. The small inset figure plots the correlation of the average experienced return series and the P/E ratio series if the weighting parameter λ is varied.

- Table VI shows that experienced stock returns also predict **expected future stock returns**. This is direct support for the beliefs channel.
- Figure IV then aggregates the experience measure and shows it lines up closely with market valuation levels. High market valuations coincide with periods when average investors have recently accumulated favorable stock experiences.
- The aggregate fit is important because the weighting parameter came from microdata rather than from fitting the stock market time series.

6 Short summary

- **Method.** The paper constructs household-specific lifetime experienced returns and estimates how those experiences affect risk taking, allowing the data to determine the weighting on recent versus distant experiences.
- **Identification.** The design exploits cross-sectional differences in lifetime macro histories while controlling for age, year, income, wealth, and household characteristics.
- **Main results.** Better experienced stock returns raise elicited risk tolerance, stock market participation, and the stock share of liquid assets; better experienced bond returns raise bond participation.
- **Mechanism.** Experienced stock returns also raise expected future stock returns, suggesting that personal experience primarily works through beliefs, though preference effects are not fully ruled out.

7 Conclusion

7.1 Core conclusion

- Lifetime macroeconomic experiences have persistent and economically meaningful effects on household financial risk taking.
- Individuals do not appear to form risk-taking decisions solely from all available historical data in an impersonal way; instead, they over-weight the return history they themselves experienced.

7.2 Economic intuition

- People seem to learn from the world they have lived through, not from the full long-run history in the textbooks.
- Recent experiences matter most because they are salient and informative to the individual, but older experiences do not fully disappear.
- As a result, generations that went through different booms and busts can systematically differ in risk taking, even after controlling for age and current aggregate conditions.

7.3 Takeaway

This paper provides one of the clearest micro-level demonstrations that personal macroeconomic experience shapes financial behavior. It links household portfolio choice, survey risk tolerance, and return expectations in a unified way. The broader implication is that asset demand, and potentially asset prices, can depend on the distribution of lived experiences across investors, not just on fundamentals and stable preferences.